Class 9 Biology **Unit- 3 Respiration and Excretion**

Model Worksheet By Rasheed Odakkal

1. Which of the following acts as a respiratory as well as an excretory ogan in humans?
Kidneys Lungs Skin Liver

2. A. Respiration is the process of taking in oxygen only.
B. Respiration is the process of releasing energy from food.
a). Both the statements are correct.
b). Statement A correct, B incorrect.
c). Statement B correct, A incorrect.
d). Both the statements are incorrect.

3. Explain the difference between aerobic and anaerobic respiration.

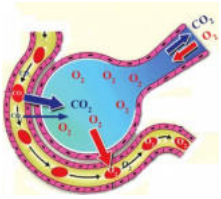
4. Draw a flowchart illustrating the path of atmospheric air from the nostrils to the alveoli.
Let's begin with,

Nostrils →

5. Ventilation process is performed by:
A. Diaphragm. B. Intercostal muscles.
a). Both A and B.
b). Only A .
c). Only B.
d). None of these.

6. The exchange of respiratory gases takes place in the _____ of lungs.

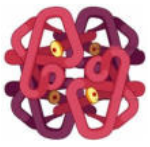
7. a). What process is being illustrated in the diagram?
b). How does the structure of the alveolus facilitate this process?
c). Which blood cells carry the oxygen and what is special in them?



8. Complete the table by inserting suitable items from the list given below:
- Exchange of gases including blood, tissue fluid and cells.
- Exchange of gases between blood and alveoli.
- Transport of gases through blood.

Alveolar exchange of gases	**Systemic exchange of gases**
• ---------	• ---------

9. a). Identify and name this molecule.
b). This molecule is seen in ----? [RBC, Plasma, WBC, Platelet]
c). Mention how this molecule is suitable for the transport of gases.



10. Complete the table by fill in the blanks suitably :

Glycolysis	------(**a**)-------
Occur in ------(**b**)-------	Occur in mitochondria
Pyruvic acid forms	------(**c**)-------
------(**d**)-------	28 ATP energy forms

11. Complete the following illustration of respiration process by adding the missing words:

Glucose + ...(**a**).... $\xrightarrow{\text{Enzymes}}$ **CO_2 +** ..... (**b**)..... **+ 30 ATP**

12. Analyse the relationships and fill in the blanks :
Amoeba : Cell membrane,
Earth worm : ---------- ?

13. Based on the following hints mention the way of gas exchange in plants:

***** Leaves ***** Stem ***** Roots.

14. Complete the table :

Excretory organ	Excretory substance
Skin	
Lungs	CO_2
Liver	Ammonia
Kidneys	

15. Protein metabolism → Ammonia → + CO2 + H2O → ------?

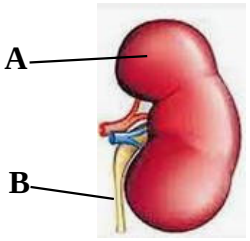
a). Complete the flowchart by adding the end product.

b). Where does this product form?

16. Observe the given figure:

a. Name the parts, labelled as A and B.

b. The broad initial part of B is known as -------- .



17. What are the functions performed by our kidneys?

18. On the basis of structure, how the afferent vessel differ from efferent vessel? How is this related with the process of ultrafiltration ?

19. Which part of the nephron is responsible for ultrafiltration?

[Renal tubule, Bowman's capsule, Collecting duct, Glomerulus]

20. Blood → Glomerular filtrate → --------- ?

21. A child is conducting an experiment by adding Benedict's reagent to one sample of urine.

a). What will be the aim of his experiment ?

b). When two smples of urine tested in a lab, bilirubin detected in one sample and calcium oxalate crystals in another one. Give your inferences about the test result.

22. - Blood from artery is passed to the unit.

- Adding heparin to prevent coagulation.
- the waste materials in the cellophane tube diffuse into the fluid.
- The purified blood is returned to the vein.

a. Name the process indicated here?

b. When does this process essential?

23. Answer in one word for each question:

a). The basic functional unit of the kidney.

b). The light coloured outer part of kidney.

c). The chief excretory substance expelled from the kidneys of reptiles and birds.

24. Make suitable pairs :

Nephridia, Contractile vacuole, Kidney, Malphigian tubule, Earthworm, Insect , Frog, Amoeba

25 . Define the following. Mention their function too.

* Hydathode * Lenticel

26. Redraw the given figure and correctly label the name of following parts :

a. Branch of aorta that carries blood in to the kidney.

b. The part where ultrafilters of nephron seen.

c. Region where urine from the filters flows into.

